

ZHYMPITY, KazakhstanWMO: **352170**Lat: **50.250N**Lon: **52.567E**Elev: **105**StdP: **14.64**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-11.7	-23.0	1.6	-17.5	-16.8	2.2	-11.2	32.1	26.5	27.5	21.1	5.0	20	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	97.6	67.9	94.4	66.8	91.2	65.7	70.5	91.0	68.9	88.7	67.4	86.4	8.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	90.4	76.5	62.0	83.4	76.1	59.9	77.3	75.7	34.4	91.0	33.0	88.8	31.8	86.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	22.1	18.8	-22.5	102.0	7.4	3.4	-27.9	104.4	-32.2	106.4	-36.4	108.3	-41.8	110.8
			-22.8	75.1	7.4	2.8	-28.1	77.1	-32.4	78.8	-36.5	80.4	-41.8	82.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	12.5	13.1	27.4	48.3	62.3	71.2	75.3	73.3	60.3	45.1	30.1	18.0
	DBStd	24.90	12.17	12.79	10.69	10.06	8.68	8.27	6.51	7.90	8.56	8.52	10.63	11.93
	HDD50	4862	1164	1033	702	148	9	0	0	14	201	599	992	992
	HDD65	8274	1629	1453	1166	505	157	32	4	19	188	616	1048	1457
	CDD50	3008	0	0	1	97	391	637	784	722	324	51	1	0
	CDD65	942	0	0	0	3	73	219	323	276	48	0	0	0
	CDH74	11922	0	0	0	67	966	2856	3997	3402	625	9	0	0
	CDH80	5653	0	0	0	8	357	1405	1978	1695	209	1	0	0
(14) Wind	WSAvg	8.2	8.9	9.3	9.3	8.9	8.2	7.8	7.3	7.3	7.4	7.8	8.0	8.6
(15) Precipitation	PrecAvg	10.2	0.7	0.6	0.7	0.8	1.1	1.1	1.1	0.6	0.7	1.1	0.9	0.7
	PrecMax	19.1	1.7	1.5	1.6	1.9	3.1	3.3	5.7	2.3	4.2	2.7	2.0	2.4
	PrecMin	5.3	0.2	0.1	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.2	0.0	0.2
	PrecStd	3.3	0.4	0.4	0.5	0.5	0.9	0.8	1.3	0.6	0.9	0.7	0.5	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	35.4	60.9	81.0	91.7	100.6	100.6	100.8	91.7	74.3	55.8	40.8
		MCWB	34.0	33.1	49.2	61.4	64.2	69.8	68.5	68.6	62.1	56.3	50.7	38.3
	2%	DB	33.3	33.7	51.2	75.1	87.7	96.8	97.0	97.0	85.6	69.5	49.8	36.3
		MCWB	31.9	32.2	44.8	58.7	63.5	67.8	67.6	67.6	61.9	53.9	46.1	34.6
	5%	DB	32.1	32.5	44.8	70.4	84.2	92.6	93.9	93.2	81.3	64.4	45.6	34.0
		MCWB	30.9	31.0	40.4	56.3	62.6	66.2	66.9	66.1	60.1	51.1	42.5	32.5
	10%	DB	29.7	30.8	39.8	65.1	79.5	88.8	90.4	89.5	76.5	59.4	42.5	32.5
		MCWB	28.3	29.2	36.4	53.0	61.1	65.2	66.1	64.9	58.3	49.1	39.5	31.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	34.1	49.5	63.2	68.4	73.4	72.6	72.4	66.7	59.1	52.3	39.2
		MCDB	34.9	34.5	58.5	76.9	84.0	91.1	91.3	92.6	86.2	71.5	54.8	40.3
	2%	WB	32.3	32.7	45.1	59.9	65.5	70.1	70.2	69.7	63.4	55.0	46.4	35.0
		MCDB	33.1	33.4	51.4	73.6	83.0	91.4	89.7	91.8	81.0	65.7	49.5	35.9
	5%	WB	30.9	31.2	40.7	56.8	63.6	68.0	68.8	67.8	61.1	52.6	42.9	32.8
		MCDB	32.0	32.4	44.7	68.1	80.9	88.2	87.8	88.6	78.1	61.9	45.5	33.8
	10%	WB	28.3	29.2	36.6	53.7	61.8	66.2	67.4	66.1	59.1	50.2	39.9	31.1
		MCDB	29.8	30.7	39.4	64.0	78.0	84.8	85.7	86.2	74.4	57.7	42.2	32.4
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	14.1	14.1	20.4	22.5	23.9	23.2	24.1	23.4	19.1	11.5	11.0
		MCDBR	8.6	10.4	19.6	26.5	27.7	28.6	27.1	27.7	30.0	27.4	14.3	8.5
		MCWBR	8.7	9.7	13.9	14.2	12.0	11.4	10.1	10.3	13.6	15.3	11.4	7.9
	5% WB	MCDBR	8.5	9.8	18.5	25.0	25.9	25.6	23.9	25.4	26.8	23.9	12.9	8.3
		MCWBR	8.7	9.5	13.5	14.5	13.1	12.2	11.0	11.3	13.5	15.1	10.9	8.0
(40) Clear-Sky Solar Irradiance	taub		0.317	0.339	0.394	0.444	0.425	0.414	0.416	0.428	0.380	0.361	0.342	0.316
	taud		2.066	2.012	2.032	2.050	2.186	2.283	2.287	2.225	2.320	2.293	2.223	2.127
	Ebn at Noon		213	242	250	252	264	268	266	255	256	236	206	194
	Edh at Noon		27	37	44	48	44	40	40	40	33	28	23	22
(44) All-Sky Solar Radiation	RadAvg		369	724	1074	1477	1915	2102	2040	1759	1267	716	329	233
	RadStd		67	84	157	124	148	161	120	109	113	86	41	60

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A